

An introductory workshop on authoring scientific publications with Quarto

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Software

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Introduction

Scientific reporting is time-consuming and iterative, often involving multiple rounds of refinement and revision in response to collaborators, editors, and reviewers (Jiang et al., 2019). Data-centric research amplifies this complexity: researchers repeatedly interrogate data, update analyses, regenerate figures, and refine narratives while meeting formatting requirements.

General-purpose word processors often hinder reproducible research by lacking integration of code, data, and outputs, introducing inefficiencies for authors and reviewers and limiting reproducibility (Haven & Ioannidis, 2025; Randall & Welser, 2018). Addressing these challenges requires not only tools but also a cultural shift toward transparency, collaboration, and accountability throughout the research lifecycle (Brownlee & Bielekova, 2021; Rasmussen et al., 2023).

Tools such as Overleaf, Living Papers, and Quarto (Allaire et al., 2025) support transparent, reproducible workflows. Quarto, a free and open-source system developed by Posit, integrates narrative prose, executable code, visualizations, formulas, and citations into a single reproducible document. Built on Pandoc (MacFarlane et al., 2025), it supports multiple programming languages, fine-grained control over code visibility, and diverse output formats (HTML, PDF, Word) while accommodating journal templates and custom designs. Beyond manuscripts, Quarto enables high-quality presentations, websites, and books, facilitating reproducible dissemination and research reuse.

This paper introduces a modular workshop designed to familiarize learners with Quarto and to strengthen project and data management skills, with an emphasis on reproducible research practices.

Statement of Need

The [workshop lesson](#) aligns with open science principles and the pedagogical philosophy of [The Carpentries](#). Existing Carpentries curricula focus on computational skills in data manipulation, analysis, and visualization, leaving a gap in instruction on publication and reproducible reporting.

Our module provides hands-on experience in creating dynamic, reproducible documents with Quarto. Learners integrate executable code, narrative text, and citations into cohesive outputs, while also learning structured file organization, naming conventions, scalable directory structures, and strategies for managing project dependencies to ensure reproducibility over time.

41 By the end of the workshop, participants are expected to integrate Quarto into their
42 workflows and apply the core principles of reproducible research effectively in their own
43 projects.

44 Target Audience

45 The workshop targets academic researchers tasked with scientific reporting, including
46 faculty, graduate students, research assistants, and postdocs. While Quarto supports Julia,
47 Python, and Observable JavaScript, this workshop focuses on R. Prior experience with R
48 and RStudio is recommended but not required. Basic familiarity with the Unix Shell and
49 Git is expected.

50 Lesson Structure

51 The lesson consists of 15 episodes in three modules:

52 Module 1 – Reproducibility and Project Organization

53 Introduces the importance of transparent workflows, core principles of reproducible research,
54 and good-enough practices for organizing projects. This module includes two episodes and
55 ensures a common foundation for all learners.

56 Module 2 – Quarto and RStudio

57 Eight episodes provide hands-on practice creating data papers in Quarto. Learners format
58 and style documents containing code, text, images, formulas, and citations, using local
59 and global (YAML) settings.

60 Module 3 – Collaboration

61 Five episodes cover collaborative workflows using Git and GitHub, dependency
62 management, reproducible environments with the renv package (Ushey & Wickham,
63 2025), and options for publishing outputs.

64 The workshop is supported by a [project example repository](#), including R scripts, data
65 files, figures, and a functional Quarto document. The curriculum is designed for flexibility,
66 allowing it to be delivered as a full workshop or as shorter, focused sessions, as outlined in
67 the instructors' notes.

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